

The written test is of objective type for 60 minutes duration. The topics for written test will be on

Programming Languages: C++, Java, Python, JavaScript

Frontend Development: HTML, CSS, JavaScript, React.js, Angular, Vue.js, Bootstrap, Tailwind CSS, Responsive UI/UX, State Management (Redux, Context API)

Show more ▾



With 2 days left, your goal is **not to learn everything**. Your goal is to **maximize your score** by revising the highest-yield topics and practicing objective questions.

Since you already have experience with **Node.js, Express, React, SQL, JavaScript, and dashboard development**, use that to your advantage rather than spending time mastering unfamiliar technologies.

Day 1 (10–12 hours)

Session 1 (3 hours) – Programming Languages

Focus on concepts common across languages rather than language-specific syntax.

C++

- Pointers & References
- Classes & Objects
- Inheritance
- Polymorphism
- Virtual Functions
- STL (Vector, Map, Set)
- Time Complexity

Java

- OOP
- Interfaces vs Abstract Class
- Exception Handling
- Collections
- JVM, JDK, JRE
- Multithreading basics

Python

- Lists, Tuples, Dictionaries
- List Comprehension
- Lambda
- Generators
- Decorators (basic)
- Exception Handling

JavaScript

- Hoisting
- Closures
- Promise
- Async/Await
- Event Loop
- == vs ===
- let vs var vs const

Session 2 (2.5 hours) – Frontend

Revise:

- HTML5 Semantic Tags
- CSS Box Model
- Flexbox
- Grid
- Media Queries
- Responsive Design
- Bootstrap Grid
- Tailwind utility classes
- React
 - Components
 - Props
 - State
 - Hooks
 - useEffect
 - useState
 - Context API
 - Redux basics
- Angular
 - Components
 - Services
 - Two-way Binding
- Vue basics

Don't go deep into Angular/Vue if you haven't used them.

Session 3 (3 hours) – Backend

Since this is your strength:

Revise

- Node.js
- Express
- Middleware
- REST API
- CRUD
- HTTP Methods
- Status Codes
- JWT Authentication
- OAuth2 (basic flow)
- Error Handling
- Logging
- MVC Architecture
- Microservices basics
- Spring Boot architecture
- Django vs Flask differences

Session 4 (2 hours) – Databases

Revise

- SQL Queries
- Joins
- Group By
- Having
- Order By
- Subqueries
- Stored Procedures
- Views
- Indexes
- Normalization
- Primary Key
- Foreign Key
- Transactions
- ACID Properties
- MongoDB basics

Practice at least

- 20 SQL MCQs

Day 2

Session 1 (3 hours)

Data Structures & Algorithms

Most important

- Arrays
- Strings
- Linked List
- Stack
- Queue
- Trees
- BST
- Heap
- Graph basics
- HashMap
- Sorting
- Binary Search
- Recursion

Know time complexities.

Session 2 (2 hours)

Operating Systems

Revise

- Process vs Thread
- Deadlock
- Paging
- Scheduling
- Virtual Memory
- Semaphore
- Mutex

Computer Networks

- OSI
- TCP/IP
- HTTP
- HTTPS
- DNS

- IP
- UDP vs TCP
- Cookies
- Sessions

Session 3 (2 hours)

Software Engineering

- SDLC
- Agile
- Scrum
- Waterfall
- Git
- Merge
- Branch
- Pull Request
- SOLID Principles
- OOP

Session 4 (2 hours)

Testing

Revise

- Unit Testing
- Integration Testing
- Regression Testing
- Black Box
- White Box
- Selenium
- Cypress
- Jest
- JUnit

Know what each tool is used for.

Session 5 (2 hours)

Cloud & DevOps

AWS

- EC2

- S3
- IAM
- RDS
- Lambda

Azure

- Virtual Machine
- Storage
- Functions

Docker

- Image
- Container
- Dockerfile
- Docker Compose

Kubernetes

- Pod
- Deployment
- Service
- Scaling

Know definitions and use cases.

Last 2 Hours Before the Exam

Do **not** study new topics.

Instead revise:

- SQL commands
- JavaScript interview questions
- React hooks
- Node.js concepts
- OOP principles
- OS scheduling
- DBMS normalization
- HTTP status codes
- Time complexity
- Git commands

Expected Weightage (Likely)

Topic

Priority

JavaScript



Topic	Priority
React	★★★★★
Node.js + Express	★★★★★
SQL + DBMS	★★★★★
Data Structures & Algorithms	★★★★★
OOP	★★★★
Operating Systems	★★★★
Computer Networks	★★★★
Git	★★★
REST APIs	★★★★
JWT Authentication	★★★
Docker	★★★
AWS	★★
Kubernetes	★★
Angular/Vue	★★
Java	★★★
Python	★★★
C++	★★★
Testing Frameworks	★★★

Exam strategy

- Spend no more than **60–90 seconds** on any single question initially. Skip difficult ones and return later.
- Eliminate obviously wrong options before guessing.
- If there is **no negative marking**, attempt every question.
- Since the test is only **60 minutes**, aim to finish one pass with about **10 minutes left** for review.

Given your recent work on a full-stack dashboard using **React, Node.js, Express, and SQL**, you're already strong in several of the highest-priority areas. Focus your remaining time on revising core CS fundamentals (DSA, OS, DBMS, Networks), JavaScript concepts, and

objective-style questions rather than trying to build new projects or learn unfamiliar frameworks from scratch. Good revision over the next two days will likely give you a better return than broad but shallow coverage of every technology listed.

     Sources

Excellent idea. A **50-MCQ mixed mock test** is much closer to what you'll face. Below is **Mock Test 1 (50 Questions)** covering the entire syllabus.

Instructions:

- Time: **60 minutes**
- Don't use Google or notes.
- Reply only with your answers (e.g., 1-B, 2-C, 3-A...).
- I'll evaluate your score, explain every wrong answer, and identify weak areas.

Programming Languages (1–10)

1. Which keyword is used to create an object in Java?

A. create

- B. object
- C. new
- D. this

2. What is the output of `typeof null` in JavaScript?

- A. null
- B. object
- C. undefined
- D. string

3. Which Python collection is immutable?

- A. List
- B. Dictionary
- C. Set
- D. Tuple

4. Which C++ concept allows the same function name with different parameters?

- A. Overriding
- B. Overloading
- C. Encapsulation
- D. Inheritance

5. Which Java keyword prevents inheritance?

- A. final
- B. static
- C. private
- D. protected

6. Which JavaScript keyword has block scope?

- A. var
- B. let
- C. Both A and B
- D. global

7. Which Python keyword is used to handle exceptions?

- A. catch
- B. exception
- C. try
- D. error

8. Which Java collection stores unique values?

- A. List
- B. ArrayList
- C. Set
- D. Queue

9. Which operator checks value and type in JavaScript?

- A. ==
- B. ===
- C. =
- D. !=

10. Which is not an OOP principle?

- A. Inheritance
- B. Polymorphism
- C. Compilation
- D. Encapsulation

Frontend Development (11–20)

11. Which HTML tag creates a hyperlink?

- A. <p>
- B.
- C. <a>
- D. <link>

12. Which CSS property changes text color?

- A. font-color
- B. text-color
- C. color
- D. background

13. Which layout system is one-dimensional?

- A. Grid
- B. Flexbox
- C. Table
- D. Float

14. React applications start rendering from which component?

- A. Main
- B. Root/App
- C. Index
- D. Home

15. Which React Hook manages component state?

- A. useMemo
- B. useEffect
- C. useState
- D. useRef

16. Redux is mainly used for

- A. Styling
- B. Routing
- C. Global State Management
- D. API calls

17. Which hook performs side effects?

- A. useState
- B. useEffect
- C. useReducer
- D. useRef

18. Tailwind CSS is

- A. Database
- B. Utility-first CSS framework
- C. JavaScript library
- D. Backend framework

19. Which HTML tag is used for forms?

- A. <table>
- B. <form>
- C. <input>
- D. <label>

20. Bootstrap uses how many columns in its grid?

- A. 8
- B. 10
- C. 12
- D. 16

Backend Development (21–30)

21. Node.js runs on

- A. JVM
- B. Chrome V8 Engine
- C. .NET
- D. Apache

22. Which Express function defines a POST endpoint?

- A. app.get()
- B. app.listen()
- C. app.post()
- D. app.use()

23. JWT stands for

- A. Java Web Token
- B. JSON Web Token
- C. JavaScript Web Token
- D. JSON Wireless Token

24. Which HTTP method updates a resource?

- A. GET
- B. POST
- C. PUT
- D. OPTIONS

25. Status code for successful request

- A. 200
- B. 301
- C. 404
- D. 500

26. Which middleware parses JSON request bodies in Express?

- A. `express.json()`
- B. `express.static()`
- C. `bodyParser.html()`
- D. `express.router()`

27. OAuth2 is mainly used for

- A. Styling
- B. Authentication/Authorization
- C. Database
- D. Caching

28. Which is a Python backend framework?

- A. React
- B. Django
- C. Angular
- D. Vue

29. REST APIs are generally

- A. Stateful
- B. Stateless
- C. Offline
- D. Encrypted only

30. Which Express function starts the server?

- A. `app.use()`
- B. `app.listen()`
- C. `app.route()`
- D. `app.get()`

Databases (31–38)

31. Which SQL clause groups rows?

- A. ORDER BY
- B. HAVING
- C. GROUP BY
- D. WHERE

32. Which JOIN returns matching records only?

- A. LEFT
- B. RIGHT
- C. INNER
- D. FULL

33. Which SQL function counts rows?

- A. SUM
- B. COUNT
- C. AVG
- D. MAX

34. Which key uniquely identifies a row?

- A. Foreign Key
- B. Composite Key
- C. Primary Key
- D. Candidate Key

35. MongoDB stores data as

- A. Tables
- B. Rows
- C. Documents
- D. Arrays only

36. Which improves query performance?

- A. Trigger
- B. View
- C. Index
- D. Cursor

37. ACID property ensuring completed transactions survive crashes

- A. Isolation
- B. Consistency
- C. Durability
- D. Atomicity

38. Which SQL clause filters grouped records?

- A. WHERE

- B. GROUP BY
- C. HAVING
- D. DISTINCT

Software Engineering (39–46)

39. Which data structure follows LIFO?

- A. Queue
- B. Stack
- C. Linked List
- D. Tree

40. Average search complexity of HashMap

- A. $O(n)$
- B. $O(\log n)$
- C. $O(1)$
- D. $O(n^2)$

41. Binary Search requires

- A. Queue
- B. Sorted array
- C. Linked List
- D. Tree

42. Which scheduling algorithm may cause starvation?

- A. FCFS
- B. Round Robin
- C. Priority Scheduling
- D. FIFO

43. Which protocol is connection-oriented?

- A. UDP
- B. TCP
- C. IP
- D. ARP

44. HTTPS uses default port

- A. 80
- B. 25
- C. 443
- D. 110

45. Git command to upload commits

- A. git fetch
- B. git pull

- C. git push
- D. git clone

46. SDLC stands for

- A. Software Development Life Cycle
- B. System Data Logic Cycle
- C. Software Design Level Code
- D. Software Data Life Cycle

Testing, Cloud & DevOps (47–50)

47. Jest is used for

- A. Database
- B. Unit Testing
- C. Networking
- D. Deployment

48. Docker packages applications into

- A. Pods
- B. Images
- C. Clusters
- D. Services

49. Smallest deployable unit in Kubernetes

- A. Cluster
- B. Node
- C. Pod
- D. Service

50. AWS S3 is mainly used for

- A. Virtual Machines
- B. Object Storage
- C. Databases
- D. Authentication

After you submit your answers, I'll:

-  Score you out of **50**
-  Explain every incorrect answer
-  Tell you your approximate readiness level
-  Generate another **50 harder MCQs** focused on placement/interview-level questions if needed.

Since you only have **2 days**, the highest-value practice is **large question banks**, not just one mock test. Aim to solve **200–300 MCQs**.

Here are **150 additional MCQs** (without answers) organized by topic so you can practice continuously.

JavaScript (20 MCQs)

1. What is closure?
2. Difference between `==` and `===` ?
3. What is hoisting?
4. What is event bubbling?
5. What is event capturing?
6. What is the event loop?
7. What is callback hell?
8. What is a Promise?
9. What does `async` return?
10. Which array method returns a new array?
11. Difference between `map()` and `forEach()`
12. Difference between `let` , `const` , `var`
13. What is destructuring?
14. What is spread operator?
15. What is rest operator?
16. What is `this` keyword?
17. What is strict mode?
18. What is prototype?
19. What is module import/export?
20. Difference between `null` and `undefined`

React (20 MCQs)

21. Purpose of Virtual DOM
22. Props vs State
23. `useState`
24. `useEffect`
25. `useMemo`
26. `useCallback`
27. `useRef`

28. Context API
29. Redux
30. React Router
31. Controlled Components
32. Uncontrolled Components
33. JSX
34. Key prop
35. Fragment
36. Conditional Rendering
37. Lifting State Up
38. React Lifecycle
39. Lazy Loading
40. Code Splitting

Node.js & Express (20 MCQs)

41. Event-driven architecture
42. Non-blocking I/O
43. Middleware
44. `express.json()`
45. `express.static()`
46. Routing
47. HTTP methods
48. Status codes
49. JWT
50. OAuth2
51. REST API principles
52. CRUD
53. MVC architecture
54. Error handling middleware
55. CORS
56. Cookies vs Sessions
57. Environment variables
58. File upload middleware
59. Logging
60. Microservices

SQL & DBMS (25 MCQs)

61. Primary Key
62. Foreign Key
63. Candidate Key
64. Composite Key

- 65. Super Key
- 66. INNER JOIN
- 67. LEFT JOIN
- 68. RIGHT JOIN
- 69. FULL JOIN
- 70. CROSS JOIN
- 71. GROUP BY
- 72. HAVING
- 73. Aggregate Functions
- 74. Subqueries
- 75. Correlated Subqueries
- 76. Views
- 77. Stored Procedures
- 78. Triggers
- 79. Indexing
- 80. Clustered Index
- 81. Normalization
- 82. ACID
- 83. Transactions
- 84. Isolation Levels
- 85. MongoDB documents

DSA (20 MCQs)

- 86. Array
- 87. Linked List
- 88. Stack
- 89. Queue
- 90. Circular Queue
- 91. Binary Tree
- 92. BST
- 93. Heap
- 94. HashMap
- 95. Graph
- 96. DFS
- 97. BFS
- 98. Binary Search
- 99. Merge Sort
- 100. Quick Sort
- 101. Bubble Sort
- 102. Insertion Sort
- 103. Selection Sort
- 104. Time Complexity
- 105. Space Complexity

Operating Systems (15 MCQs)

- 106. Process
- 107. Thread
- 108. Scheduling
- 109. FCFS
- 110. Round Robin
- 111. SJF
- 112. Deadlock
- 113. Semaphore
- 114. Mutex
- 115. Paging
- 116. Segmentation
- 117. Virtual Memory
- 118. Cache Memory
- 119. Context Switching
- 120. Starvation

Computer Networks (15 MCQs)

- 121. OSI Model
- 122. TCP/IP
- 123. HTTP
- 124. HTTPS
- 125. DNS
- 126. DHCP
- 127. IP Address
- 128. MAC Address
- 129. TCP vs UDP
- 130. SMTP
- 131. FTP
- 132. SSH
- 133. Firewall
- 134. Cookies
- 135. Sessions

Git (10 MCQs)

- 136. git init
- 137. git clone
- 138. git add
- 139. git commit
- 140. git push

- 141.** git pull
- 142.** git fetch
- 143.** git merge
- 144.** git rebase
- 145.** git stash

Testing (10 MCQs)

- 146.** Unit Testing
- 147.** Integration Testing
- 148.** System Testing
- 149.** Smoke Testing
- 150.** Regression Testing
- 151.** Jest
- 152.** JUnit
- 153.** Selenium
- 154.** Cypress
- 155.** Mocking

Cloud & DevOps (20 MCQs)

- 156.** AWS EC2
- 157.** AWS S3
- 158.** AWS RDS
- 159.** AWS Lambda
- 160.** IAM
- 161.** Azure VM
- 162.** Azure Functions
- 163.** Docker Image
- 164.** Docker Container
- 165.** Dockerfile
- 166.** Docker Compose
- 167.** Kubernetes Pod
- 168.** Deployment
- 169.** Service
- 170.** ReplicaSet
- 171.** Cluster
- 172.** Namespace
- 173.** CI/CD
- 174.** Jenkins
- 175.** GitHub Actions

Exam-focused tip

Given your syllabus and the technologies you've recently worked with (React, Node.js, Express, SQL), prioritize:

1. JavaScript
2. React
3. Node.js & Express
4. SQL/DBMS
5. DSA
6. OOP
7. Operating Systems
8. Computer Networks

These topics typically account for the majority of software developer objective tests.

If you want a **real company-style practice set**, I can also generate **300 objective MCQs with four options and answers at the end**, similar to placement exams from companies like TCS, Infosys, Accenture, Cognizant, Capgemini, Deloitte, and product-based companies.



[more mcqs you give](#)

Absolutely! Since you're preparing for a **60-minute software developer written test**, here are **50 new objective MCQs (Questions 51–100)** with four options each. **No answers are included** so you can test yourself. After you attempt them, I'll evaluate your answers.

Programming Languages (51–60)

51. Which of the following is NOT a primitive data type in JavaScript?

- A. String
- B. Boolean
- C. Object
- D. Number

52. Which keyword is used to stop a loop in C++?

- A. continue
- B. stop
- C. break
- D. exit

53. Which Java keyword is used to achieve abstraction?

- A. abstract
- B. virtual

- C. override
- D. extends

54. Which Python function returns the length of a list?

- A. size()
- B. count()
- C. len()
- D. length()

55. Which JavaScript method converts JSON text into an object?

- A. JSON.stringify()
- B. JSON.parse()
- C. JSON.object()
- D. JSON.convert()

56. Which access modifier makes a member accessible only within the same class?

- A. public
- B. protected
- C. private
- D. default

57. Which operator is used for exponentiation in Python?

- A. ^
- B. **
- C. %
- D. //

58. Which loop executes at least once?

- A. for
- B. while
- C. do-while
- D. foreach

59. Which Java collection maintains insertion order?

- A. HashSet
- B. TreeSet
- C. LinkedHashSet
- D. PriorityQueue

60. Which keyword refers to the current object in Java?

- A. self
- B. current
- C. this
- D. super

Frontend (61–70)

61. Which HTML element is used to display an image?

- A. <picture>
- B.
- C. <image>
- D. <src>

62. Which CSS property controls the space outside an element?

- A. padding
- B. border
- C. margin
- D. spacing

63. Which CSS property changes font size?

- A. text-size
- B. font-style
- C. font-size
- D. size

64. React components should always start with

- A. Small letters
- B. Numbers
- C. Capital letters
- D. Symbols

65. Which hook is used to access Context API?

- A. useEffect
- B. useContext
- C. useState
- D. useMemo

66. Which Bootstrap class creates a primary button?

- A. btn-blue
- B. btn-primary
- C. button-primary
- D. primary-btn

67. Tailwind class for display:flex

- A. flex
- B. display-flex
- C. d-flex
- D. flexbox

68. Which attribute uniquely identifies a list item in React?

- A. id
- B. key

- C. name
- D. value

69. Which HTML input type hides typed characters?

- A. hidden
- B. secure
- C. password
- D. text

70. Which CSS unit is relative to the root font size?

- A. px
- B. em
- C. rem
- D. %

Backend (71–80)

71. Which Node.js module is used for file operations?

- A. fs
- B. path
- C. os
- D. net

72. Express middleware executes

- A. Before route handler
- B. After response only
- C. Only on errors
- D. Never

73. Which HTTP method is idempotent?

- A. POST
- B. GET
- C. PATCH
- D. CONNECT

74. Which status code means Unauthorized?

- A. 200
- B. 401
- C. 403
- D. 500

75. JWT consists of how many parts?

- A. 2
- B. 3
- C. 4
- D. 5

76. Which Express method serves static files?

- A. app.static()
- B. express.static()
- C. app.public()
- D. express.public()

77. Which package is commonly used for JWT authentication in Node.js?

- A. express-jwt
- B. jsonwebtoken
- C. jwt-auth
- D. node-token

78. Which database is commonly used with the MERN stack?

- A. Oracle
- B. MongoDB
- C. SQLite
- D. MariaDB

79. Which HTTP method deletes a resource?

- A. REMOVE
- B. DELETE
- C. ERASE
- D. CLEAR

80. Which architecture separates application into small independent services?

- A. MVC
- B. Monolithic
- C. Microservices
- D. Layered

SQL & DBMS (81–90)

81. Which SQL command removes all rows but keeps the table structure?

- A. DROP
- B. DELETE
- C. TRUNCATE
- D. REMOVE

82. Which constraint prevents duplicate values?

- A. PRIMARY KEY
- B. UNIQUE
- C. FOREIGN KEY
- D. CHECK

83. Which SQL operator searches a pattern?

- A. MATCH

- B. SEARCH
- C. LIKE
- D. FIND

84. Which clause sorts query results?

- A. SORT
- B. GROUP BY
- C. ORDER BY
- D. HAVING

85. Which join returns all rows from the left table?

- A. RIGHT JOIN
- B. INNER JOIN
- C. LEFT JOIN
- D. CROSS JOIN

86. MongoDB stores data in

- A. Tables
- B. Documents
- C. Rows
- D. Cells

87. Which SQL function returns the highest value?

- A. TOP()
- B. HIGH()
- C. MAX()
- D. UPPER()

88. Which normal form removes transitive dependency?

- A. 1NF
- B. 2NF
- C. 3NF
- D. BCNF

89. Which command permanently removes a table?

- A. DELETE
- B. REMOVE
- C. DROP
- D. TRUNCATE

90. Which SQL keyword removes duplicate records?

- A. UNIQUE
- B. DISTINCT
- C. GROUP
- D. FILTER

DSA, OS, CN, Git, Testing, Cloud (91–100)

91. Which data structure is best for recursion?

- A. Queue
- B. Stack
- C. Heap
- D. Tree

92. Time complexity of binary search is

- A. $O(n)$
- B. $O(\log n)$
- C. $O(n^2)$
- D. $O(1)$

93. Which scheduling algorithm gives every process equal CPU time?

- A. FCFS
- B. SJF
- C. Round Robin
- D. Priority

94. Which protocol transfers web pages?

- A. FTP
- B. SMTP
- C. HTTP
- D. SSH

95. Which Git command creates a local repository?

- A. git start
- B. git init
- C. git create
- D. git new

96. Selenium is mainly used for

- A. Database testing
- B. UI Automation
- C. API Gateway
- D. Docker

97. Docker images are created using

- A. Dockerfile
- B. Dockercompose
- C. Podfile
- D. Manifest

98. Kubernetes is mainly used for

- A. Database management
- B. Container orchestration

- C. Unit testing
- D. Version control

99. Which AWS service provides virtual servers?

- A. S3
- B. EC2
- C. Lambda
- D. IAM

100. Which AWS service is serverless?

- A. EC2
- B. Lambda
- C. RDS
- D. EBS

These 50 questions are representative of what you might encounter in a software developer written test.

If you're aiming for thorough preparation, I can also create:

- **MCQs 101–200** (medium difficulty)
- **MCQs 201–300** (advanced/company-style)
- **100 scenario-based MCQs** (code snippets and debugging)
- **100 SQL query-based MCQs**
- **100 JavaScript & React output-based MCQs**

This would give you around **600 practice questions** across your syllabus.

